MANAGE THE DYNAMIC OCCLUSION OF IMPLANT RESTORATIONS IN A DIGITAL WORKFLOW

Complete implantology treatments must be both esthetic and functional, and digital technologies make it possible to anticipate the final esthetic result and plan the implants in an ideal position. The Modjaw device allows the same degree of anticipation. After having carried out a functional recording of the patient, the healthcare team determines a prosthetic plan with an ideal static and dynamic occlusion. Guided surgery and immediate loading make it possible to transfer this project precisely into the mouth and to obtain an aesthetic and functional result in accordance with the planning.

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Links of interest
The author declares that he has no ties of interest.

Bibliographic referencing
Set A. Managing dynamic occlusion of implant restorations in a digital flow. CLINIC 2022;43(411):00-00.

The digital revolution in the dental profession has made it possible to plan cases very precisely and to approach full-arch implantology in an efficient and effective manner [1].

It is therefore usual to make a photographic assessment, allowing to plan the aesthetics of the smile in two dimensions (2D) with the help of a smile design tool (Smile Design). It is also usual to merge the optical impressions of the arches and the 2D Smile Design to obtain a 3D *wax-up*, a faithful reproduction of the 2D draft. By matching this 3D design with the DICOM data from the *cone beam*, it is possible to plan the implants in ideal position and to transpose this planning into the mouth using guided surgery. Finally, due to the mechanical properties of PMMA resins, it has recently become possible to attach the temporary bridge to the implants even before the end of the procedure. This approach allows immediate rehabilitation of the aesthetics of the smile. However, as it stands, it is quite difficult to record, respect, and reproduce the pre-existing function *(Figure 1)*.

This requires:

– to integrate an additional device to the digital chain, the Modjaw, allowing to record the existing function to analyze it and to replay it in our CAD software;

– ensure an exact repositioning of the prosthesis in order to allow the patient to regain the pre-existing static and dynamic occlusion in the mouth. **The purpose of this article is to present the integration of the Modjaw system in complete implantology treatments.** To do so, we will review the concepts of occlusion management in complete implant rehabilitation, the functioning of the Modjaw device, and, finally, we will summarize its interest through a clinical case.

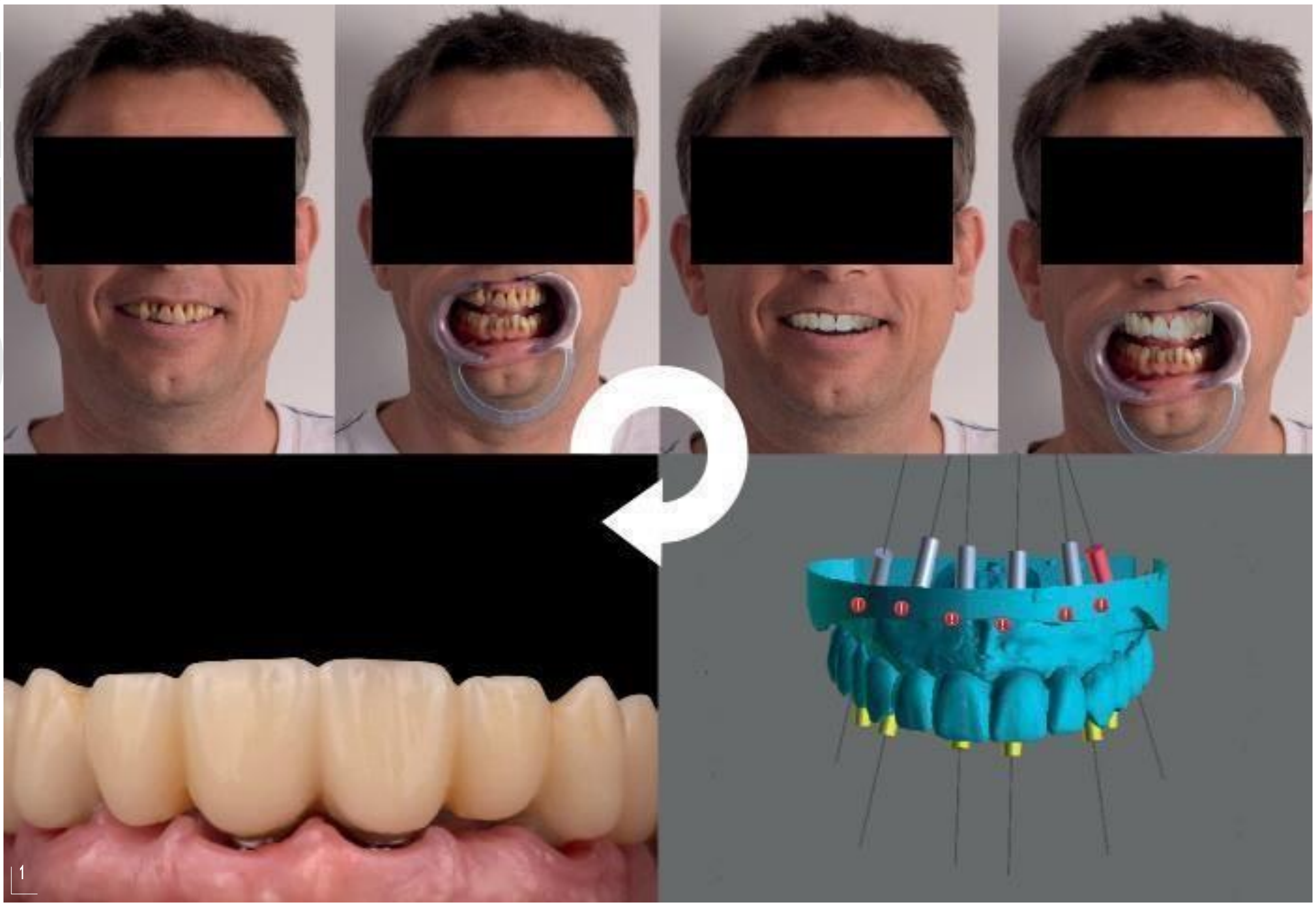


Figure 1
Intraoperative loading protocol.

OCCLUSION MANAGEMENT OF COMPLETE IMPLANT-SUPPORTED RESTORATIONS

Intermaxillary relationship

Choosing the intermaxillary relationship (IMR) means finding a reproducible and physiological position to reconstruct the occlusion [2].

During the clinical examination of a dentate patient, 3 situations could be present:

– the patient has a stable and reproducible maximum intercuspation (MI) and functional temporomandibular joints (TMJ). The patient's MI will then be chosen as the reference position for the prosthetic rehabilitation;

– the patient has a stable and reproducible occlusion in MI but impaired TMJ function. The chosen IMR will be the therapeutic position obtained following the treatment of his joint disorder;

– the patient does not have a stable and reproducible occlusion in MI. The IMR will then be the occlusion in centric relation (ORC) obtained by manipulation/accompaniment of his mandible, or the physiological position of the muscles at rest (PPMR) obtained with the help of a TENS, a jig, or a stop (neuro-muscular position obtained during the closing after deprogramming of the engrams).

Choice of vertical dimension

As in the case of a full denture, maintaining or modifying the vertical dimension of occlusion (VDO) is an essential choice from both an esthetic and functional point of view [3]. From an aesthetic point of view, it is a question of balancing the 3 levels of the face. From a functional point of view, it ensures good masticatory and phonetic function and avoids joint problems.

If the patient is edentulous, the VD will be recorded at the same time as the IMR, using an occlusal base. If the patient is still dentate, we record the existing VDO and the IMR, which allows the models to be placed in the articulator, and then the height of the pin will eventually be reassessed. This classic and analog approach often poses many concerns because the hinge axis of an articulator does not correspond exactly to the real and physiological placement of the condyles during an increase in VD. We will see in the following section that the Modjaw is a tool of choice to perform a VD augmentation using the patient's actual physiological data.

Static occlusion

The static occlusion of complete restorations on implants is like that of a natural arch. The aim is to obtain a cuspid/fossa ratio, with balanced contacts on the posterior teeth and slightly less marked contacts on the anterior teeth.
Let's keep in mind that this MI occlusion is a position found only during swallowing and mastication, that is to say approximately 25 to 40 minutes per day.

Dynamic occlusion

A distinction is made here between occlusal-prosthetic aspects, which are intended to protect our restorations and the TMJ, and functional dynamic occlusion (which promotes mastication), which is intended to provide comfort and guide the mandible in daily life.

OCCLUSAL-PROSTHETIC ASPECTS [4].

In the majority of cases, we will use the mutually protected occlusion (MPO). In this concept, the protrusion is guided by the anterior guide, which ensures that the posterior teeth are in disocclusion during this movement, particularly in the case of distal extensions of implant-supported restorations. The laterotrusions are guided by the canine on the working side (canine function) or by the canine and the adjacent teeth (group function). In this concept, guiding the working side leads to the disocclusion of the non-working side teeth. This is also particularly important in the case of possible distal extensions.
If the prosthetic rehabilitation is performed in conjunction with an implant-supported full denture or prosthesis, the concept of bilaterally balanced occlusion (BBO) or balanced occlusion is chosen. Protrusion is guided by balanced anterior and posterior contacts. The laterotrusion is supported by the working and non-working sides, so as not to destabilize the prosthesis of the antagonist arch. The choice of design will always be determined by the less stable arch.

PHYSIOLOGICAL ASPECT AND MASTICATION

It is a question, here, of adjusting the occlusion in order to ensure the load of the chewing movement [5].

The enamel bridge of the maxillary first molar has a major role in the management of the in and out cycles during the masticatory movement. This movement remains very difficult to regulate, but we will see in the rest of this article that the Modjaw makes it possible to record it and thus to reconstruct a masticatory function close to the initial one.

THE MODJAW DEVICE

The Modjaw is a device allowing :
– to record the real mandibular kinematics;
– to carry out a precise occlusal analysis;
– to record the patient's axiographic data.

The device consists of:
– a cart supporting a computer with a touch screen and a camera;
– a mandibular butterfly, a frontal tracker(Tiara) and a stylus, all three carrying sensors to be visualized by the camera *(figure 2)*. The general principle is to merge the patient's over-facial impressions, on his own arches, in order to use the patient's joint as an articulator [6].

Figure 2
Modjaw device.

Preliminary steps

Before making the recordings, a few preliminary steps are necessary:
– import the *.stl*, *.ply* or *.obj* digital files in occlusion, from the optical or physical prints of the patient ;
– select 4 reference points on the mandibular impression, to be able to find them in the mouth to *match* the mandibular file to the real mandibular arch of the patient. We also indicate to the system, the mandibular inter-incisal point, to allow it find the reference point in the mouth.

Figure 3 *Matching points of the mandibular arch.*

Figure 4 *Mounting sensors Modjaw.*

Figure 5 *Condyles and subnasal point to be matched.*

Figure 6 *Matching points of the mandibular arch in the mouth.*

This allows the calculation of the motion amplitudes *(Figure 3)*;

– to fix the different sensors. First the mandibular butterfly, which is positioned on a fork, itself fixed to the mandibular teeth with a bisacryl resin, then the Tiara *(figure 4)*;

– calibrate all the sensors of the fork, the tiara and stylus;

– indicate to the system the position of the patient's condyles using the stylus. This is a fundamental point because the real hinge axis of the patient will be used and not an arbitrary axis as it is the case with an articulator *(figure 5)*;

– locate the 4 points of the mandibular file on the arch to use the patient as a physiological articulator *(figure 6)*.

The device is then ready to be used and we can start the functional recordings.

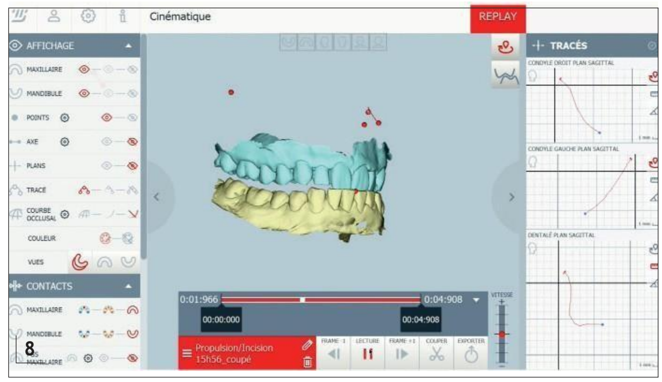
Routine recordings and clinical interests

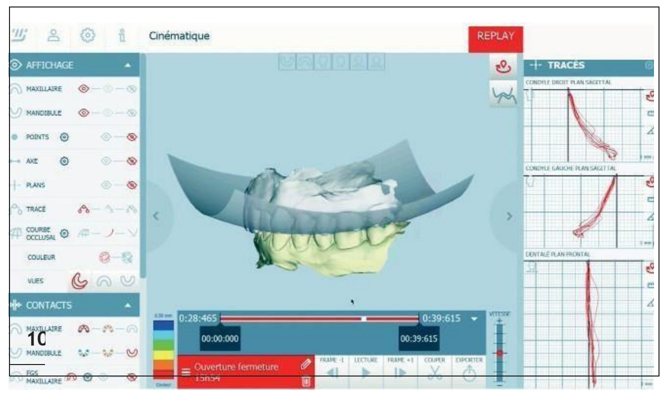
• Opening/closing *(figure 7)*: the patient is asked to open to the maximum opening several times, then to reduce the amplitude of the movement but to accelerate the repetitions until a series of "tap-taps". The amplitude of the movement can also be measured and is between 50.7 mm ± 7 mm. The rapid tap-tap movement ensures that the patient returns to the MI position in a reproducible manner. If this is the case (and in the absence of temporomandibular dysfunctions), we can consider choosing the MI position as the reference position for the complete implant rehabilitation.

• Protrusion *(figure 8)*: the recording of the movement starts in MIP then the patient protrudes while keeping the dental contact until his most extreme anterior position. The analysis of the function is done by replaying the movement, especially in slow motion, or even frame by frame. The quality of the anterior guide, the distribution of the guiding surfaces on the different incisors, as well as the presence of possible posterior interferences are then assessed. The system also automatically calculates the condylar slopes and the amplitude of the movement (7.3 to 9.1 mm on average) can be measured.

• Lateralities *(figure 9)*: the recording of the movement starts in MI and then the patient slides in maximum laterality while keeping the dental contact. The analysis of the function is done in the same way and we can clearly objectify if the patient presents a canine or group function and if he presents non-working interferences. The system also automatically calculates the Benett angles and we can measure the amplitude of movement (9 to 10 mm on average).

• Centeric relationship or muscular rest position: if we choose not to use the patient's MI, Modjaw is





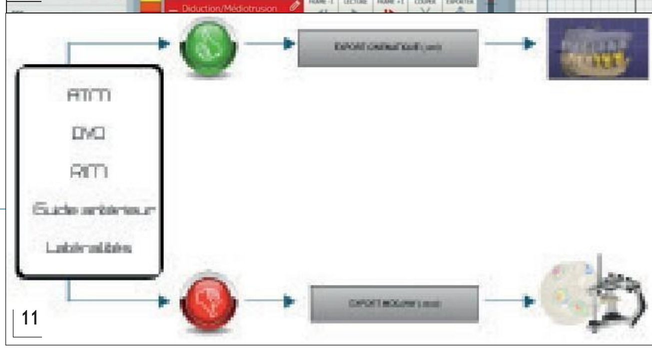


Figure 7 *Opening/closing movement.*
Figure 8 *Propulsion.*
Figure 9 *Right laterality.* **Figure 10** *Manson's sphere.* **Figure 11** *.xml and .mod exports.*

very precise and determines the new inter-maxillary relationship. The manipulation towards the centric relation or the position following a neuro-muscular deprogramming is recorded and the movement can be replayed to fix this new IMR. The dental technician can then use this new position in his CAD software.

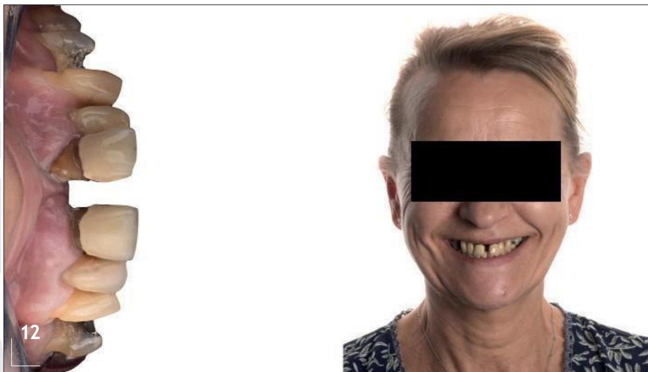
• Chewing: this is probably the most negative movement recorded by the device. The patient is asked to chew a piece of apple or chewing gum and the actual chewing movement is recorded. The analysis consists of the appreciation of the vertical and transversal amplitudes of his chewing cycles. The prosthodontist can use the recording of this movement to design occlusal tables adapted to the existing kinematics.

Clinical and prosthetic use of recordings

• The dome: this dome gives the first indications on the position of the patient's teeth in relation to the Spee and Wilson curves *(Figure 10)*.

• Data export *(figure 11)*: the use of the records in the prosthetic laboratory requires their export from the system to be imported into a CAD software such as Exocad.

Two types of export are possible. The first one in *.xml* only concerns the movements that will be replayed by the dental technician to adjust the dynamic occlusion of our patient. We use this export when we do not fundamentally modify the patient's occlusion. The second usable export concerns the complete export of the consultation (export *.mod*). In this case, the dental technician also receives the axiographic data and can program a virtual articulator. We use this export when we modify the IMR or the VDO of the patient for example. The reading of this *.mod* file requires a Modjaw software version on the laboratory side.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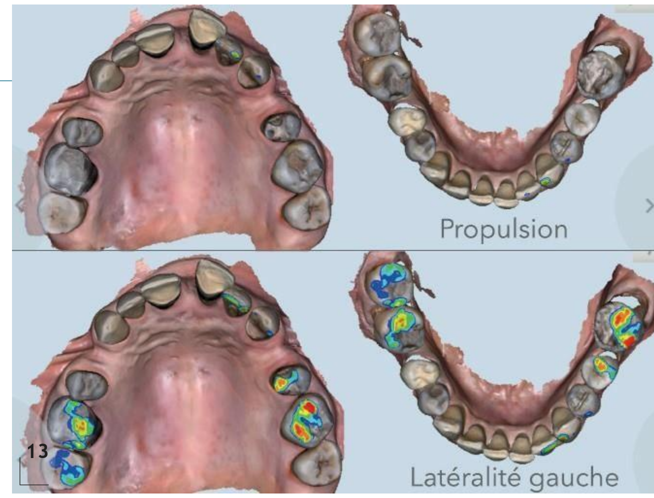

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Figure 12 *Initial situation.*
Figure 13 *Dynamic occlusion.*
Figure 14 *Smilecloud aesthetic project.*
Figure 15 *4D Wax-up.*

CLINICAL CASE

A 58-year-old female patient, with no previous medical history, presented for aesthetic and functional reasons. She had stage 4 and grade C periodontitis, causing numerous terminal mobilities, real aesthetic prejudice, and difficulties in eating *(figure 12)*.

It was decided to treat her periodontitis in the mandible and to offer her a complete fixed prosthesis on 6 implants in the maxilla. The 6 implants will be placed in immediate extraction-implantation and a provisional prosthesis will be placed intraoperatively with the help of digital planning [7].

Data collection

A "digitization" consultation allows us to collect the digital data necessary for planning. It consists of a photographic *charting*, a *cone beam*, optical impressions of both arches and occlusion as well as a functional registration with the Modjaw.

Analysis of the Modjaw recordings show a stable and reproducible MI but a disturbed function in protrusion and laterality. Indeed, there are many non-working contacts in laterality and posterior interference in protrusion. Moreover, only the 22 provides anterior guidance in this case. Restoring optimal function is therefore an essential objective for the durability of the future restoration *(Figure 13)*.

Aesthetic and prosthetic project

All the photographs are imported on the Smilecloud platform and the 2D design of the smile is realized [8] *(figure 14)*. We also import the files from the optical impressions and the *.xml* export of the Modjaw data to order the digital file of the prosthetic project on the platform. This file is made with the help of the teeth library used during the 2D design and is adjusted with the help of the data recorded with the Modjaw. This project thus goes beyond 3D by integrating function: we call it a 4D *wax-up (figure 15)*.

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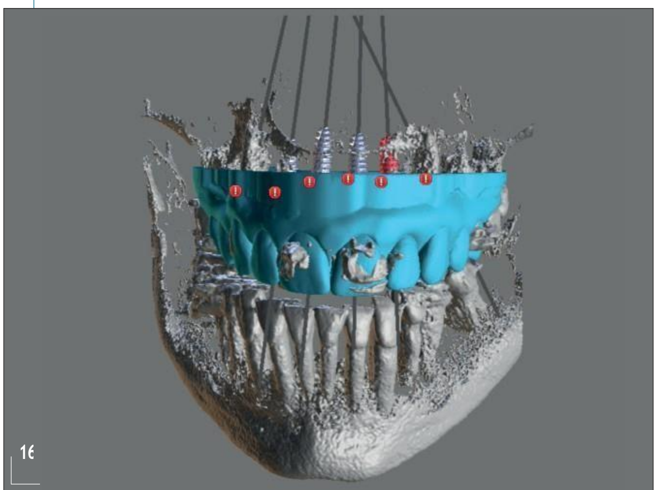

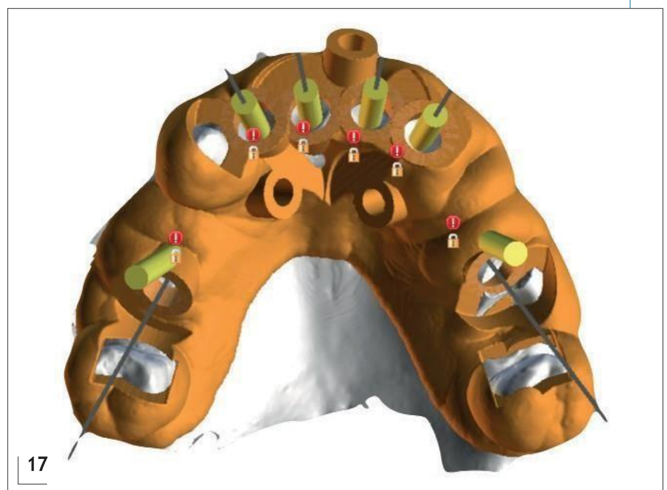

Figure 16 *Implant planning.*
Figure 17 *Surgical guide.*

We have the possibility to validate the dynamic occlusion of this project by importing it into the Modjaw to replay the recordings with this new occlusion.
Once this 4D *wax-up has been* made and validated, it becomes the reference that will guide each step of the treatment: the implants are planned according to this project and the provisional prosthesis will be a carbon copy; if the patient validates this provisional prosthesis from an aesthetic and functional point of view, the prosthesis in use will also be made by a digital copy of this 4D *wax-up.*

Implant planning

The planning of this case was carried out using Blue Sky Plan software.
After the alignment of the initial model, the 4D *wax-up* is imported and merged onto the previous files.
The implants are then virtually positioned in an ideal position from a prosthetic and anatomical point of view *(Figure 16)*.
The initial model is then virtually edentulous, trying to keep a maximum of teeth (13, 17 and 27 in this case). This will allow us to *design* a guide with mixed mucosal and dental support, which is much easier to reposition precisely than a guide with mucosal support alone [9].
This guide has palatal and vestibular guide tubes to receive stabilization screws *(Figure 17)*.

Temporary prosthesis

By exporting the 4D *wax-up*, the position of the implants and the 2 palatal guide tubes, the dental technician has the necessary information to create the provisional prosthesis. The provisional prosthesis has perforations opposite the future implants and is attached to a palatal spacer with the two palatal guide tubes that we have on the surgical guide *(Fig. 18)*.

Surgery

After local anesthesia, the teeth were removed, except for 13, 17 and 27. The guide was carefully positioned, and stabilized with osteosynthesis screws and the In-Kone® (Global D) implants were placed through the guide. The guide was then removed and 13 was extracted. It was decided to retain 17 and 27 during healing, to allow the patient to retain some of her proprioception *(Fig. 19)*.
The provisional is repositioned and screwed into the two palatal holes. It is then attached to the titanium provisional sheaths, which are themselves screwed onto MUA-type abutments *(Figures 20 and 21)*. The palatal spacer is then cut, and the bridge is carefully polished and screwed onto the implants.

Result

From an aesthetic point of view, the result is in accordance with the planning and the patient is delighted *(Figure 22)*.
To evaluate the obtained function, we performed a new Modjaw analysis.

We can see that the anterior guide is restored and distributed over 3 incisors instead of one initially. There are no more posterior interferences. The absence of discomfort related to the modification of the anterior guide will of course be verified during this transitional phase. The left laterality is taken care of by 23 and 24 (semi-group function), without any non-working interference *(figure 23)*.

The right laterality is taken care of by the 13 (canine function), without non-working interference, exactly as planned before the operation.

The chewing cycles are large and regular.

The functional result of the treatment is therefore in accordance with the planning.

Prosthesis for use

The provisional prosthesis being validated from an esthetic and functional point of view, it was decided to produce the customary prosthesis by copying the 4D *wax-up*.

In order not to change the *design*, the prosthesis is machined in zirconia and only the false gingiva is made manually.

First, the *design of* the prosthesis is printed in resin to ensure that it conforms to the provisional prosthesis both aesthetically and occlusally *(Figure 24)*.

After validation, the prosthesis of use is carried out then screwed(Laboratoire Drumez in Annequin) *(figure 25)*.

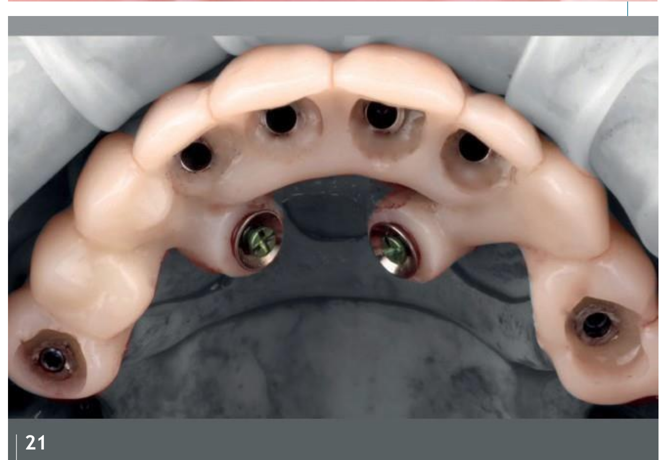

Figure 18 *Temporary prosthesis.*
Figure 19 *Surgical guide stabilized in the mouth.*
Figure 20 *Implant emergence through the provisional.*
Figure 21 *Bonding of the provisional titanium sleeves to the provisional bridge.*

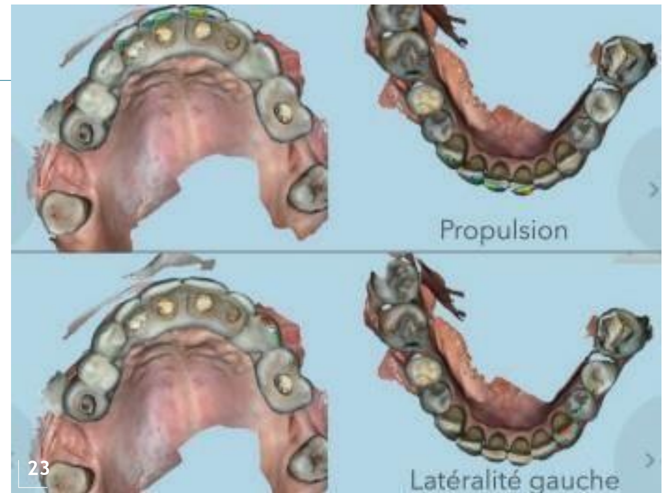

Figure 22 *Aesthetic result of intraoperative loading.*
Figure 23 *Functional result of the intraoperative loading.*
Figure 24 *Printed fitting model.*
Figure 25 *Prosthesis in use.*

CONCLUSION

Managing the dynamic occlusion of our implant restorations in a fully digital workflow is now a reality.
We understand more than ever the preponderant role of the prosthetic project in this type of rehabilitation because, once this project is established, we only work by copying it at each stage of the treatment.

This implies a great deal of rigor in the planning before the procedure and great precision in the repositioning of the guide and the provisional in the mouth.
Only if these two conditions are we met can rehabilitate the aesthetics and function of our patients in a single intervention.

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